

MIKHAYLOVKA, Kazakhstan

WMO: 298020

Lat: 53.8394N

Lon: 76.5411E

Elev: 114

StdP: 99.96

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.2	-32.4	-37.7	0.1	-35.1	-35.0	0.1	-32.3	9.7	-6.0	8.7	-6.7	1.9	330	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	31.7	18.8	29.9	18.1	28.0	17.5	20.2	27.7	19.5	26.6	18.7	25.6	3.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.1	22.1	17.2	12.4	21.6	16.2	11.7	21.3	58.7	27.7	56.1	26.6	53.6	25.5	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.9	7.0	-36.1	35.0	3.4	2.0	-38.5	36.4	-40.5	37.6	-42.4	38.7	-44.9	40.1		
(5)			DB	-36.5	22.1	4.1	1.6	-39.4	23.3	-41.8	24.2	-44.2	25.2	-47.2	26.4	(4)
			WB													(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-18.4	-16.4	-6.7	5.3	12.5	19.0	19.9	17.8	11.7	3.9	-5.9	-14.7		
	DBStd	14.95	8.54	7.37	6.15	6.03	5.19	4.30	3.49	3.77	4.74	5.02	7.24	8.10		
	HDD10.0	3814	881	739	517	165	36	2	0	0	34	198	477	766		
	HDD18.3	5998	1139	973	775	392	191	43	24	56	206	448	727	1024		
	CDD10.0	1050	0	0	0	23	112	271	308	242	85	9	0	0		
	CDD18.3	192	0	0	0	0	9	62	73	40	7	0	0	0		
	CDH23.3	2136	0	0	0	13	164	668	718	464	107	1	0	0		
	CDH26.7	698	0	0	0	2	41	229	244	151	31	0	0	0		
(14) Wind	WSAvg	3.3	3.1	3.2	3.6	3.8	3.7	3.1	2.9	2.9	3.0	3.2	3.5	3.4		
(15) Precipitation	PrecAvg	302	15	10	11	15	29	39	56	43	30	28	21	16		
	PrecMax	535	60	24	38	44	139	126	147	148	85	62	53	37		
	PrecMin	140	3	0	1	1	7	1	2	5	9	2	1	1		
	PrecStd	90	13	7	8	11	27	25	37	33	18	15	12	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	1.1	8.6	25.3	31.0	33.6	34.0	33.6	30.6	20.8	8.2	1.6		
		MCWB	0.1	0.0	4.7	14.4	16.5	18.9	20.6	19.7	17.0	11.5	5.9	0.5		
	2%	DB	-1.3	-0.2	4.5	21.4	27.3	31.2	31.5	30.6	26.4	17.4	5.9	0.5		
		MCWB	-2.3	-1.1	2.3	12.7	14.9	18.3	19.4	18.4	14.8	10.1	3.9	-0.3		
	5%	DB	-3.4	-2.5	2.6	18.3	25.0	29.5	29.4	28.1	23.1	14.0	3.8	-1.0		
		MCWB	-4.2	-3.3	1.0	10.8	14.2	17.8	18.7	17.4	13.8	8.3	2.2	-1.8		
	10%	DB	-5.5	-5.2	1.4	14.9	22.2	27.3	27.3	25.9	20.0	11.2	2.1	-3.0		
		MCWB	-6.1	-5.9	0.2	9.0	13.0	17.3	18.3	16.7	12.5	7.1	0.8	-3.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	0.2	4.9	14.9	17.5	20.6	21.7	20.6	18.0	12.2	6.2	0.8		
		MCDB	1.0	0.9	8.1	23.7	28.6	29.2	29.6	28.8	28.9	18.4	7.9	1.3		
	2%	WB	-2.1	-1.1	2.5	13.2	15.8	19.7	20.5	19.3	15.9	10.6	4.1	-0.2		
		MCDB	-1.4	-0.3	4.2	19.9	25.1	28.1	27.5	26.7	23.8	16.6	5.7	0.4		
	5%	WB	-4.1	-3.3	1.3	11.3	14.8	18.9	19.8	18.5	14.5	8.9	2.4	-1.7		
		MCDB	-3.3	-2.6	2.4	17.5	23.2	26.4	26.5	25.7	21.9	13.1	3.6	-1.0		
	10%	WB	-6.1	-5.9	0.2	9.3	13.6	18.0	19.1	17.6	13.0	7.4	0.9	-3.8		
		MCDB	-5.5	-5.2	1.4	14.3	21.0	25.4	25.5	24.5	19.2	10.9	1.9	-3.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.1	8.3	10.6	12.5	11.7	10.9	11.6	10.9	8.6	6.7	7.9		
		MCDBR	7.6	7.1	7.3	15.5	16.2	14.4	14.1	15.6	15.8	13.8	6.3	8.2		
	5% WB	MCWBR	7.3	6.8	6.0	8.1	7.2	5.4	5.0	6.1	7.0	7.6	5.0	8.2		
		MCWBR	7.7	7.1	7.4	14.9	14.3	12.3	11.5	12.9	14.2	12.1	6.1	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.292	0.322	0.381	0.405	0.407	0.413	0.436	0.423	0.370	0.351	0.330	0.295		
	taud		2.039	1.993	1.929	2.155	2.208	2.274	2.213	2.218	2.319	2.256	2.112	2.004		
	Ebn at Noon		626	735	769	818	839	837	807	793	794	711	574	524		
	Edh at Noon		73	108	145	133	133	126	132	124	98	83	69	63		
(44) All-Sky Solar Radiation	RadAvg		1.11	2.22	3.78	4.70	5.81	6.21	5.61	4.73	3.41	1.77	0.85	0.72		
	RadStd		0.08	0.09	0.27	0.36	0.47	0.38	0.38	0.36	0.32	0.22	0.11	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page